Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In Bernoulli's theorem which of the following is conserved?

(A) Mass

(B) Linear momentum

(C) Energy

(D) Angular momentum

Q.2

A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth $4y$ from the top. When the tank is completely filled with water, the quantities of water flowing out per second from both holes are the same. Then, $\mathrm{R} =$

(A) $\frac{L}{\sqrt{2\pi}}$

(B) $2\pi\mathrm{L}$

(C) L

(D) $\frac{L}{2\pi}$

Q.3

A wind with speed $40\ \mathrm{m/s}$ blows parallel to the roof of a house. The area of the roof is $250\ \mathrm{m}^2$. Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind on the roof and the direction of the force will be $\left(\rho_{\text{air}} = 1.2\ \mathrm{kg/m}^3\right)$

(A) $4.8 \times 10^5\ \mathrm{N}$, upwards

(B) $2.4 \times 10^5\ \mathrm{N}$, upwards

(C) $2.4 \times 10^5\ \mathrm{N}$, downwards

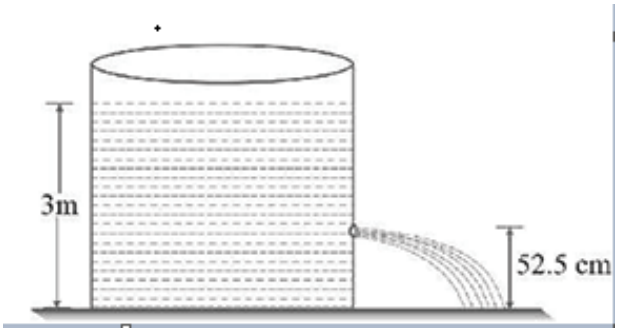
(D) $4.8 \times 10^5\ \mathrm{N}$, downwards

Q.4

Water is filled in a cylindrical container to a height of $3\ \mathrm{m}$. The ratio of the cross-sectional area of the orifice

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and the beaker is 0.1. The square of the speed of the liquid coming out from the orifice is $\left(g=10 \mathrm{~m} / \mathrm{s}^2\right)$



(A) $50 \mathrm{~m}^2 / \mathrm{s}^2$

(B) $50.5 \mathrm{~m}^2 / \mathrm{s}^2$

(C) $51 \mathrm{~m}^2 / \mathrm{s}^2$

(D) $52 \mathrm{~m}^2 / \mathrm{s}^2$

Q.5

Water is flowing through a horizontal tube having cross-sectional areas of its two ends being A and A' such that the ratio A / A' is 5 . If the pressure difference of water between the two ends is $3 \times 10^5 \mathrm{~N} \mathrm{~m}^{-2}$, the velocity of water with which it enters the tube will be (neglect gravity effects)

(A) $5 \mathrm{~m} \mathrm{~s}^{-1}$

(B) $10 \mathrm{~m} \mathrm{~s}^{-1}$

(C) $25 \mathrm{~m} \mathrm{~s}^{-1}$

(D) $50 \sqrt{10} \mathrm{~ms}^{-1}$

Q.6

A wide vessel with a small hole at the bottom is filled with water (density ρ_1, height h_1) and kerosene (density ρ_2, height h_2). Neglecting viscosity effects, the speed with which water flows out is :

(A) $\left[2 g\left(h_1+h_2\right)\right]^{1 / 2}$

(B) $\left[2 g\left(h_1 \rho_1+h_2 \rho_2\right)\right]^{1 / 2}$

(C) $\left[2 \mathrm{~g}\left(h_1+h_2\left(\rho_2 / \rho_1\right)\right)\right]^{1 / 2}$

(D) $\left[2 \mathrm{~g}\left(h_1+h_2\left(\rho_1 / \rho_2\right)\right)\right]^{1 / 2}$

Q.7

The cylindrical tube of a spray pump has radius, R, one end of which has n fine holes, each of radius r. If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes is :

(A) $\frac{\mathrm{VR}^2}{\mathrm{nr}^2}$

(B) $\frac{\mathrm{VR}^2}{\mathrm{n}^3 \mathrm{r}^2}$

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(C) $\frac{V^2R}{nr}$

(D) $\frac{VR^2}{n^2r^2}$

Q.8

Water from a tap emerges vertically downwards with an initial speed of $1.0\ ms^{-1}$. The cross-sectional area of the tap is $10^{-4}\ m^2$. Assume that the pressure is constant throughout
the stream of water and that the flow is steady. The cross-sectional area of the stream $0.15\ m$ below the tap is (take $g = 10\ ms^{-2}$)

(A) $5.0 \times 10^{-4}\ m^2$

(B) $1.0 \times 10^{-5}\ m^2$

(C) $5.0 \times 10^{-5}\ m^2$

(D) $20 \times 10^{-5}\ m^2$

Q.9

The volume of an air bubble is doubled as it rises from the bottom of a lake to its surface.
The atmospheric pressure is $75\ cm$ of mercury and the ratio of the density of mercury to that of lake water is $40/3$. What is the depth of the lake?

(A) $10\ m$

(B) $15\ m$

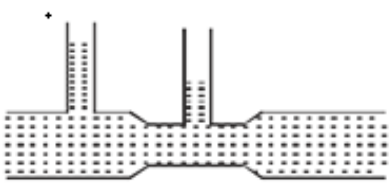
(C) $20\ m$

(D) $25\ m$

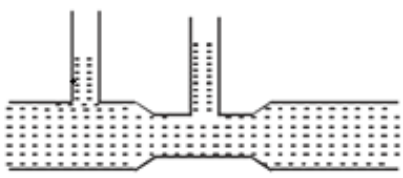
Q.10

For a fluid which is flowing steadily, the level in the vertical tubes is best represented by

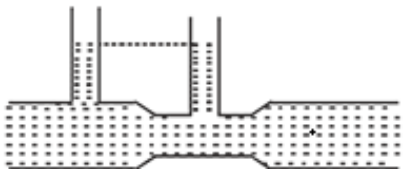
(A)



(B)



(C)



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(D)



Q.11

Water is filled in a cylindrical container to a height of $3\ \mathrm{m}$. The ratio of the cross-sectional area of the orifice and the beaker is 0.1. The square of the speed of the liquid coming out from the orifice is $\left(g = 10\ \mathrm{m/s^2}\right)$

(A) $50\ \mathrm{m/s^2}$

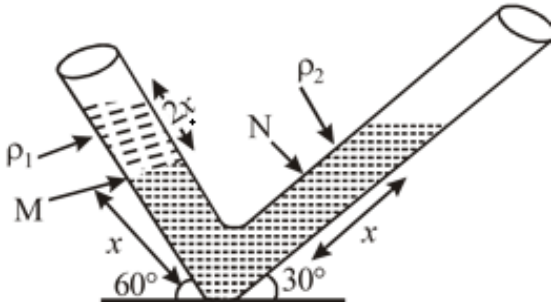
(B) $50.5\ \mathrm{m/s^2}$

(C) $51\ \mathrm{m/s^2}$

(D) $52\ \mathrm{m/s^2}$

Q.12

A liquid of density p_1 and ρ_2 stand in the bent tube as shown. Then



(A) $P_M \neq P_N$

(B) $p_2 = 2\sqrt{3}p_1$

(C) $\rho_1 = 2\sqrt{3}p_2$

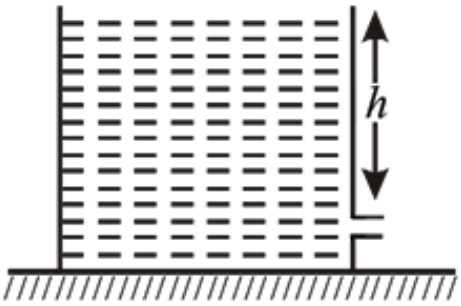
(D) $P_N = P_X$

Q.13

In the figure shown, a light container is kept on a horizontal rough surface of coefficient of friction $\mu = \frac{Sh}{V}$. A very small hole of area S is made at depth h. Water of volume V is filled in the container. The friction is not sufficient to keep the container at rest. The acceleration of the

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container initially is



(A) $\frac{V}{Sh}g$

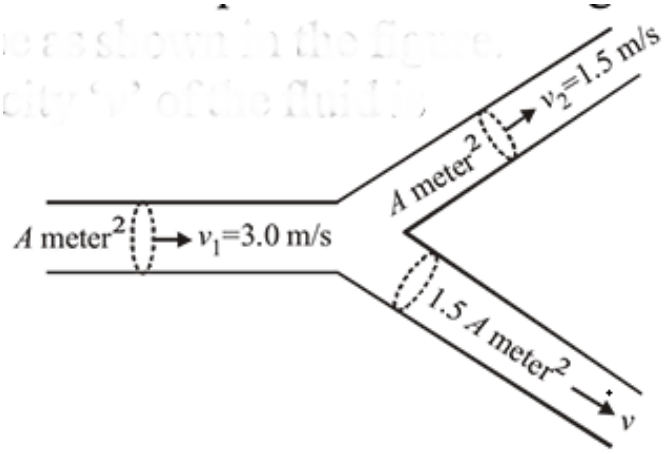
(B) g

(C) zero

(D) $\frac{Sh}{V}g$

Q.14

An incompressible liquid flows through a horizontal tube as shown in the figure. Then the velocity ' v ' of the fluid is



(A) $3.0\ \mathrm{m/s}$

(B) $1.5\ \mathrm{m/s}$

(C) $1.0\ \mathrm{m/s}$

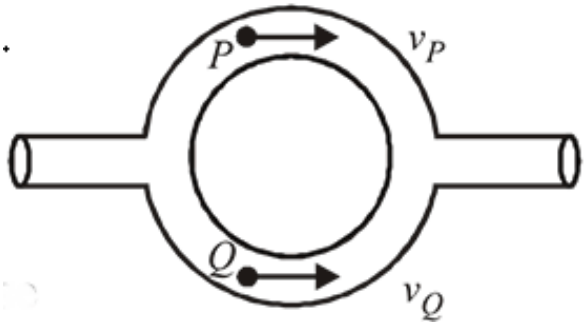
(D) $2.25\ \mathrm{m/s}$

Q.15

Figure shows a liquid flowing through a tube at the rate of $0.1\ \mathrm{m^3/s}$. The tube is branched into two semicircular tubes of cross - sectional area $\mathrm{A/3}$ and $2\ \mathrm{A/3}$. The velocity of liquid at Q is (the cross-section

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of the main tube is $\mathrm{A} = 10^{-2}\ \mathrm{m}^2$ and $v_p = 20\ \mathrm{m/s}$)



(A) $5\ \mathrm{m/s}$

(B) $30\ \mathrm{m/s}$

(C) $35\ \mathrm{m/s}$

(D) None of these

Answers & Solutions

Q.1 Answer:

Energy

Solution:

In Bernoulli's theorem only law of conservation of energy is obeyed.

Q.2 Answer:

$\frac{L}{\sqrt{2\pi}}$

Solution:

$L^2 + \sqrt{2g} = \pi R^2 \sqrt{2z(4y)}$

or $\mathrm{L}^2 = 2\pi\mathrm{R}^2$ **or** $R = \frac{L}{\sqrt{2\pi}}$

Q.3 Answer:

$2.4 \times 10^5\ \mathrm{N}$, upwards

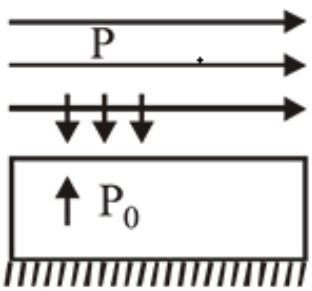
Solution:

According to Bernoulli's theorem,

$\mathrm{F} = \Delta\mathrm{PA} = \frac{1}{2}\rho \mathrm{v}^2\ \mathrm{A}$ $\uparrow \mathrm{P}_0$

$= \frac{1}{2} \times 1.2 \times 40 \times 40 \times 250 = 2.4 \times 10^5\ \mathrm{N}$

(upwards)



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Q.4 Answer:

$50\ \mathrm{m^2/s^2}$

Solution:

If v_2 is the required speed then $A_1v_1 = A_2v_2$
or $v_1 = \frac{A_2}{A_1}v_2 = 0.1v_2$
From Bernoulli's equation, we have $P_a + \frac{1}{2}\rho v_1^2 + \rho g(3 - 0.525) = P_a + \frac{1}{2}\rho v_2^2 + 0 \ldots$
After solving above equations, we get $v_2^2 = 50\ \mathrm{m^2/s^2}$

Q.5 Answer:

$5\ \mathrm{m\ s^{-1}}$

Solution:

According to Bernoulli's theorem $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$
From question, $P_1 - P_2 = 3 \times 10^5$, $\frac{A_1}{A_2} = 5$
According to equation of continuity $A_1v_1 = A_2v_2$ **or,** $\frac{A_1}{A_2} = \frac{v_2}{v_1} = 5 \Rightarrow v_2 = 5v_1$
From equation (i) $P_1 - P_2 = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right)$
or $3 \times 10^5 = \frac{1}{2} \times 1000\left(25v_1^2 - v_1^2\right)$
$\Rightarrow \quad 600 = 6v_1 \times 4v_1 \Rightarrow v_1^2 = 25$
$\therefore \quad v_1 = 5\ \mathrm{m/s}$

Q.6 Answer:

$[2\ \mathrm{g}\ (h_1 + h_2\ (\rho_2/\rho_1))]^{1/2}$

Solution:

$$P_a + \frac{1}{2}\rho_1 v_1^2 + 0$$
$$= Pa + \frac{1}{2}\rho_2 v_2^2 + (\rho_1 g h_1 + \rho_2 g h_2)$$
$$As v_2 << v_1, \quad \therefore v_1 = \sqrt{2g\left(h_1 + h_2\frac{\rho_2}{\rho_1}\right)}$$

Q.7 Answer:

$\frac{\mathrm{VR^2}}{\mathrm{nr^2}}$

Solution:

Inflow rate of volume of the liquid = Outflow rate of volume of the liquid $\pi \mathrm{R}^2\ \mathrm{V} = \mathrm{n}\pi\mathrm{r}^2(\mathrm{v}) \Rightarrow$
$\mathrm{v} = \frac{\pi \mathrm{R}^2\ \mathrm{V}}{\mathrm{n}\pi\mathrm{r}^2} = \frac{\mathrm{VR}^2}{\mathrm{nr}^2}$

Q.8 Answer:

$5.0 \times 10^{-5}\ \mathrm{m^2}$

Solution:

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The equation of continuity of flow is $u_1a_1 = u_2a_1$ where $u_1 = 1.0\ \mathrm{ms}^{-1}, a_1 = 10^{-4}m^2, u_2 =$ velocity of

stream at $\mathrm{h} = 0.15\ \mathrm{m}$ below the tap The value of v_2 is given by

$u_2^2 = u_1^2 + 2gh = 1.0 + 2 \times 10 \times 0.15 = 4.0$

or $u_2 = 2.0\ \mathrm{ms}^{-1}$

Nom $a_2 = \frac{u_1a_1}{u_2} = \frac{1.0\times10^{-4}}{2.0}$

$= 5.0 \times 10^{-5}\ \mathrm{m}^2$

Q.9 Answer:

$10\ \mathrm{m}$

Solution:

Let V_1 be the volume of the bubble and P_1 the pressure on it when it is at the bottom of the lake. If h is the depth of the lake, we have

$P_1 = P_0 + hpg$ **(here $p =$ density of wrater)**

where P_0 is the atmospheric pressure. If V_2 is the volume of the bubble and P_2 the pressure on it when it reaches the surface of the lake, then

$\mathbf{P}_2 = \mathbf{P_0}$

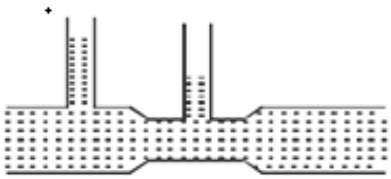
$\mathrm{V}_2 = 2\ \mathrm{V}_1$ **(given)**

From Boyle's law, $\mathrm{P}_1\ \mathrm{V}_1 = \mathrm{P}_2\ \mathrm{V}_2$

or $(\mathrm{P}_0 + \mathrm{h}\rho g)\ V_1 = \mathrm{P}_0 \times 2V_1$**, we have**

$\mathrm{h} = \frac{\mathrm{P}_0}{\mathrm{Pg}} = \frac{0.75\times40/3\times9.8}{1\times9.8} - 10\ \mathrm{m}$

Q.10 Answer:



Solution:

From continuity equation, velocity at crosssection (1) is more than that at cross-section (2). Hence $\mathrm{P}_1 < \mathrm{P}_2$.

Q.11 Answer:

$50\ \mathrm{m/s}^2$

Solution:

Let $A =$ cross-section of tank

$\mathrm{a} =$ cross-section hole

$V =$ velocity with which level decreases

$v =$ velocity of effux

From equation of continuity

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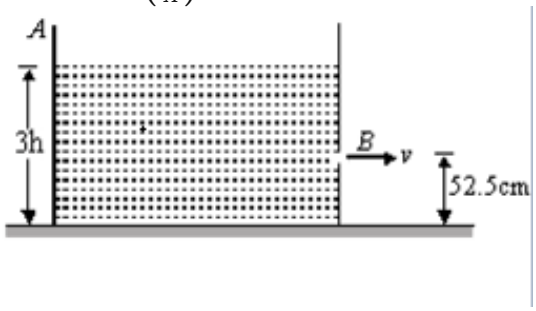
$\alpha v = AV \Rightarrow \gamma = \frac{\sigma v}{A}$

By using Bernoulli's theorem for energy per unit volume

Energy per unit volume at point A = Energy per unit volume at point B

$P + Fd + \frac{1}{2}p^a = P + 0 + \frac{1}{2}\rho v^2$

$\Rightarrow v^2 = \frac{2gh}{1-\left(\frac{a}{A}\right)^2} = \frac{2\times 10x(3-0.525)}{1-(0.1)^2} = 50(\pi/8ec)^2$



Q.12 Answer:

$p_2 = 2\sqrt{3}p_1$

Solution:

$2x \sin 60 p_1 g = x \sin 30 p_2 g$

$2\frac{\sqrt{3}}{2}\rho_1 = \frac{1}{2}\rho_2$

$p_2 = 2\sqrt{3}\rho_1$

Q.13 Answer:

$\frac{Sh}{V}g$

Solution:

Let the density of water be ρ, then the force by escaping liquid on container $= \rho S(\sqrt{2gh})^2$

- Acceleration of container $a = \frac{2\rho Sgh - \mu\rho Vg}{\rho V} = \left(\frac{2Sh}{V} - \mu\right)g$

Now, $\mu = \frac{Sh}{V} \quad \therefore a = \frac{Sh}{V}g$

Q.14 Answer:

$1.0\ \mathrm{m/s}$

Solution:

$A_1v_1 = A_2v_2 + A_3v_3$

$A \times 3 = A \times 1.5 + 1.5\ \mathrm{A} \times \mathrm{v_3} \Rightarrow v_3 = 1\ \mathrm{m/s}$

Q.15 Answer:

$5\ \mathrm{m/s}$

Solution:

From the equation of continuity, Rate $= r = \frac{A}{3}V_p + \left(A - \frac{A}{3}\right)V_Q$

$\mathrm{V_P} + 2\ \mathrm{V_Q} = \frac{3\mathrm{r}}{\mathrm{A}} = \frac{3\times 0.1}{10^{-2}} = 30\ \mathrm{m/s}$

As $V_P = 20\ \mathrm{m/s}$**, so** $\mathrm{V_Q} = 5\ \mathrm{m/s}$

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